

修士論文

# Microlocal Morse Theory and Truncated Microsupport

大柴寿浩

2026.3

北海道大学大学院理学院  
数学専攻博士前期課程



# Microlocal Morse Theory and Truncated Microsupport

大柴寿浩

## 目次

|     |                                                             |   |
|-----|-------------------------------------------------------------|---|
| 1   | Preface                                                     | 1 |
| 2   | Notations and Reviews                                       | 1 |
| 3   | Truncated Microsupport                                      | 2 |
| 3.1 | Functorial Properties . . . . .                             | 2 |
| 4   | Lemmas from [KS90]                                          | 2 |
| 5   | Results                                                     | 3 |
| 6   | Truncated Noncharacteristic Deformation                     | 6 |
| 7   | Relative Microlocal Morse Theory via Truncated Microsupport | 8 |

## 1 Preface

In this thesis, We study microlocal Morse theory, which is presented, for instance, in [GKS12]. However, we replace the notion of microsupports with that of truncated microsupprts, introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KFS03]. In [KFS03], they do not study the functorial property of truncated microsupport of direct images of non proper morphisms. We prove this functorial property under non proper direct images.

## 2 Notations and Reviews

We consider a commutative unital ring  $\mathbf{k}$  of finite global dimension (e.g.  $\mathbf{Z}$  or a field). Let  $X$  be a real analytic manifold. We denote the cotangent bundle of  $X$  by  $\pi: T^*X \rightarrow X$ . We denote by  $D^b(\mathbf{k}_X)$  the bounded derived category of sheaves of  $\mathbf{k}$ -modules on  $X$ .

Let us denote  $Z$  a locally closed set of  $X$  and  $i: Z \hookrightarrow X$  an inclusion. We use the following functors:

$$R\Gamma_Z(\cdot) := Ri_*i^!, \quad (\cdot)_Z := Ri_!i^{-1}.$$

## Geometrical Notations

Let  $M$  be a manifold. We denote by  $\pi: T^*M \rightarrow M$  the cotangent bundle of  $M$ . If  $f: M \rightarrow N$  is a morphism of manifolds, then we have the natural diagram.

$$\begin{array}{ccccc} TM & \xrightarrow{f'} & M \times_N TN & \xrightarrow{f_\tau} & TN \\ & \searrow \tau_M & \downarrow N & \square & \downarrow \tau_N \\ & & M & \xrightarrow{f} & N, \end{array} \quad \begin{array}{ccccc} T^*M & \xleftarrow{f_d} & M \times_N T^*N & \xrightarrow{f_\pi} & T^*N \\ & \searrow \pi_M & \downarrow N & \square & \downarrow \pi_N \\ & & M & \xrightarrow{f} & N. \end{array}$$

In this diagram,  $f_\tau$  and  $f_\pi$  denote the natural projections.  $f'$  is defined by  $(x; v) \mapsto (x; df_x v)$ , and  $f_d$  is induced by the transposition of  $f'$ .

## 3 Truncated Microsupport

**Definition 3.1.** Let  $F \in D^b(\mathbf{k}_X)$ . The closed conic subset  $SS_k(F)$  of  $T^*X$  is defined by:  $p \notin SS_k(F)$  if and only if there exists an open conic neighborhood  $U$  of  $p$  such that for any  $x \in \pi(U)$  and for any real-valued  $C^1$ -function  $\varphi$  defined on a neighborhood of  $x$  such that  $\varphi(x) = 0$ ,  $d\varphi(x) \in U$ , one has

$$(3.1) \quad H_{\{\varphi \geq 0\}}^j(F)_x = 0 \quad \text{for any } j \leq k.$$

### 3.1 Functorial Properties

**Proposition 3.2.** Let  $f: X \rightarrow Y$  be a morphism of real analytic manifolds and let  $F \in D^b(\mathbf{k}_X)$  such that  $f$  is proper on  $\text{supp}(F)$ . Then for any  $k \in \mathbf{Z}$ ,

$$(3.2) \quad SS_k(Rf_* F) \subset f_\pi f_d^{-1}(SS_k(F)).$$

The equality holds in case  $f$  is a closed embedding.

## 4 Lemmas from [KS90]

Consider a projective system of abelian groups  $(M_n, \rho_{n,p})_n$  indexed by  $\mathbf{N}$ . One says that this system satisfies the Mittag-Leffler condition (ML for short) if for any  $n \in \mathbf{N}$ , the decreasing sequence  $(\rho_{n,p}(M_p))_p$  of subgroups of  $M_n$  is stationary.

For a projective system of abelian groups  $(M_n)_n$ , we set

$$M_\infty := \varprojlim_n M_n.$$

Consider a projective system of complexes

$$(4.1) \quad E_n^\bullet: \cdots \rightarrow M_n^{j-1} \rightarrow M_n^j \rightarrow M_n^{j+1} \rightarrow \cdots,$$

and projective limit

$$(4.2) \quad E_\infty^\bullet: \dots \rightarrow M_\infty^{j-1} \rightarrow M_\infty^j \rightarrow M_\infty^{j+1} \rightarrow \dots$$

Denote by

$$(4.3) \quad \Phi_k: H^k(E_\infty^\bullet) \rightarrow \varprojlim_n H^k(E_n^\bullet)$$

the natural morphism.

**Proposition 4.1** ([KS90, Proposition 1.12.4]). Assume that for each  $j \in \mathbf{Z}$ , the system  $(M_n^j)_n$  satisfies the ML condition. Then

- (a) for each  $k \in \mathbf{Z}$ , the map  $\Phi_k$  is surjective,
- (b) if moreover, for a given  $i$  the system  $(H^{i-1}(E_n^\bullet))_n$  satisfies the ML condition, then  $\Phi_i$  is bijective.

The following is called Kashiwara's lemma.

**Proposition 4.2** ([KS90, Proposition 1.12.6]). Let  $(X_s, \rho_{s,t}: X_t \rightarrow X_s)_{(s,t) \in \mathbf{R}, (s \leq t)}$  be a projective system of sets indexed by  $\mathbf{R}$ . Assume that for each  $s \in \mathbf{R}$ , the canonical maps

$$\lambda_s: X_s \rightarrow \varprojlim_{r < s} X_r$$

and

$$\mu_s: \varinjlim_{t > s} X_t \rightarrow X_s$$

are both injective (resp. surjective). Then all the maps  $\rho_{s_0, s_1}$  ( $s_0 \leq s_1$ ) are injective (resp. surjective).

## ML procedure for sheaves

**Proposition 4.3** ([KS90, Proposition 2.7.1]). Let  $X$  be a topological space and let  $F \in \mathbf{D}^+(\mathbf{Z}_X)$ . Let  $(U_n)_{n \in \mathbf{N}}$  be an increasing sequence of open subsets of  $X$ ,  $(Z_n)_{n \in \mathbf{N}}$  a decreasing sequence of closed subsets of  $X$ . Set  $U := \bigcup_n U_n$ ,  $Z := \bigcap_n Z_n$ .

- (i) For any  $j$ , the natural map  $\varphi_j: H_Z^j(U; F) \rightarrow \varprojlim_n H_{Z_n}^j(U_n; F)$  is surjective.
- (ii) Assume that for a given  $j$ , the projective system  $(H_{Z_n}^{j-1}(U_n; F))_n$  satisfies the ML condition. Then  $\varphi_j$  is bijective.
- (iii) Let now  $(X_n)_{n \in \mathbf{N}}$  be an increasing family of subsets of  $X$  satisfying  $X = \bigcup_n X_n$  and  $X_n \subset \text{Int}(X_{n+1})$  for all  $n$ . Assume that for a given  $j$ , the projective system  $(H^{j-1}(X_n; F))_n$  satisfies the ML condition. Then the natural map  $H^j(X; F) \rightarrow \varprojlim_n H^j(X_n; F)$  is bijective.

## 5 Results

In this section, let  $M$  be a manifold.\*<sup>1</sup>

---

\*<sup>1</sup>Can  $M$  be of class  $C^\infty$ ?

**Theorem 5.1.** Let  $F$  in  $D^b(\mathbf{k}_M)$ , and  $\psi: M \rightarrow \mathbf{R}$  be a real-valued function of class  $C^1$ , proper on  $\text{supp}(F)$ . Let  $a < b$  be real numbers. We assume  $d\psi(x) \notin \text{SS}_k(F)$  for  $a \leq \psi(x) < b$ . For  $t \in \mathbf{R}$ , we set  $M_t := \psi^{-1}(]-\infty, t[)$ . Then, the restriction morphisms

$$H^j(M_b; F) \rightarrow H^j(M_a; F)$$

are isomorphic for each  $j < k$ .<sup>\*2</sup>

**Lemma 5.2.** Let  $-\infty < a < b < +\infty$  be real numbers and  $G \in D^b(\mathbf{R})$ . Let  $k$  be an integer. If

$$\left( H^j_{[t, +\infty[}(G) \right)_t \cong 0 \quad \text{for all } t \in [a, b[, \text{ and all } j \leq k.$$

Then one has

$$H^j(]-\infty, b[; G) \xrightarrow{\sim} H^j(]-\infty, a[; G)$$

for all  $j < k$ .<sup>\*3</sup>

**Proof.** Let  $t \in [a, b[$ . We will denote by  $I_t$  the open interval  $]-\infty, t[$  and denote by  $Z_t$  the closed interval  $]-\infty, t[ (= \overline{I_t})$ . By applying the functor  $(\cdot)_{\overline{I_t}}$  to the distinguished triangle

$$\text{R}\Gamma_{\mathbf{R}-I_t}(G) \rightarrow G \rightarrow \text{R}\Gamma_{I_t}(G) \xrightarrow{+1} \quad \text{in } D^b(\mathbf{k}_{\mathbf{R}}),$$

we have

$$(\text{R}\Gamma_{[t, +\infty[}(G))_{\{t\}} \rightarrow G_{Z_t} \rightarrow \text{R}\Gamma_{I_t}(G) \xrightarrow{+1} \quad \text{in } D^b(\mathbf{k}_{\mathbf{R}}).$$

Applying the functor  $\text{R}\Gamma(\mathbf{R}; \cdot)$  to this, we have

$$(5.1) \quad (\text{R}\Gamma_{[t, +\infty[}(G))_t \rightarrow \text{R}\Gamma(Z_t; G) \rightarrow \text{R}\Gamma(I_t; G) \xrightarrow{+1} \quad \text{in } D^b(\mathbf{k}).$$

Then consider the following conditions for  $j \in \mathbf{Z}$ ,  $s \in [a, b[$ , and  $t \in ]a, b[$ :

$$\begin{aligned} (a)_s^j & \quad \varinjlim_{t>s} H^j(I_t; G) \xrightarrow{\sim} H^j(I_s; G), \\ (b)_t^j & \quad \varprojlim_{s<t} H^j(I_t; G) \xleftarrow{\sim} H^j(I_t; G). \end{aligned}$$

By hypothesis, it follows from (5.1) that  $H^j(Z_t; G) \xrightarrow{\sim} H^j(I_t; G)$  for  $t \in [a, b[$ .<sup>\*4</sup> Since  $\varinjlim_{t>s} H^j(I_t; G) \cong H^j(Z_s; G)$ , (a)<sub>s</sub><sup>j</sup> holds for any  $j < k$  and  $s \in [a, b[$ .<sup>\*5</sup> Moreover, since we assume  $G$  is a bounded complex,

---

<sup>\*2</sup>For  $j = k$ , this morphism might be injective.

<sup>\*3</sup>For  $k$ , this morphism might be injective. (not written in this draft.)

<sup>\*4</sup>Similarly, we have injections  $H^k(Z_t; G) \hookrightarrow H^k(I_t; G)$  for  $t \in [a, b[$ .

<sup>\*5</sup>Similarly,

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(Z_s; G)$$

gives injections

$$\varinjlim_{t>s} H^k(I_t; G) \hookrightarrow H^k(I_s; G)$$

for all  $s \in [a, b[$ .

$(b)_t^j$  holds for  $j \ll 0$  and for all  $t \in ]a, b]$ . Let us denote such  $j$  by  $j_0$ . Let us argue by induction on  $j$ . Assume  $(b)_t^j$  holds for all  $t \in ]a, b[$  and all  $j \leq j_0$ . Applying Proposition 4.2, we find the isomorphisms

$$(5.2) \quad H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j \leq j_0 \text{ and all } a < s \leq t < b.$$

Let  $G \xrightarrow{\sim} G^\bullet$  be a flabby resolution. Let  $t \in ]a, b]$ . Consider a complex

$$E_n^\bullet = \Gamma(I_{t-\frac{1}{n}}; G^\bullet).$$

Since  $G^\bullet$  is flabby, we have  $E_n^j \cong E_\infty^j$  and the projective system  $(E_n^j)_n$  satisfies the ML condition for all  $j \in \mathbf{Z}$ .

By (5.2), the projective system  $(H^{j_0}(E_n^\bullet))_n$  satisfies the ML condition (because we have  $a < t - \frac{1}{n} < t$  for  $n \gg 0$ ). Applying Proposition 4.1 to  $(E_n^\bullet)_n$ , it follows that  $(b)_t^j$  holds for  $j_0 + 1$  and the induction proceeds. More precisely, we have

$$\begin{aligned} H^{j_0+1}(I_t; G^\bullet) &\xrightarrow{\sim} \varprojlim_{\Phi_{j_0+1}} H^{j_0+1}(I_{t-\frac{1}{n}}; G^\bullet) \\ &\cong \varprojlim_n H^{j_0+1}(I_{t-\frac{1}{n}}; G) \\ &\cong \varprojlim_{s < t} H^{j_0+1}(I_s; G). \end{aligned}$$

Again by Proposition 4.2, we get the isomorphisms

$$H^j(I_s; G) \xrightarrow{\sim} H^j(I_t; G) \quad \text{for all } j < k \text{ and all } a < s \leq t \leq b,$$

and injections

$$H^j(I_s; G) \hookrightarrow H^j(I_t; G)$$

for  $j = k$  and for all  $a < s \leq t \leq b$ .

Finally, using (5.1) and the hypothesis, we have the isomorphisms

$$H^j(I_a; G) \xleftarrow{\sim} H^j(Z_a; G) \xrightarrow{\sim} H^j(I_s; G) \quad \text{for all } j < k \text{ and all } a < s \leq b.$$

Similarly, for  $j = k$ , we have

$$H^k(I_b; G) \cong \varinjlim_{t > s} H^k(I_t; G) \hookrightarrow H^k(I_s; G) \hookrightarrow H^k(I_a; G).$$

□

**Proof of Theorem 5.1.** First, we have

$$H^j(M_t; F) \cong H^j \mathrm{R}\Gamma([- \infty, t[; \mathrm{R}\psi_* F)$$

for  $t \in I$ . Since we assume that  $d\psi(x) \notin \mathrm{SS}_k(F)$  for  $a \leq \psi(x) < b$ , and by (3.2), the formula  $\mathrm{SS}_k(\mathrm{R}\psi_* F) \subset \psi_\pi \psi_d^{-1} \mathrm{SS}_k(F)$  holds. Hence we have

$$(H^j \mathrm{R}\Gamma_{[\psi(x), +\infty[}(\mathrm{R}\psi_* F))_{\psi(x)} = 0$$

for each  $j \leq k$ . Then we can apply Lemma 5.2 and have

$$H^j([- \infty, b[; \mathrm{R}\psi_* F) \xrightarrow{\sim} H^j([- \infty, a[; \mathrm{R}\psi_* F)$$

for all  $j < k$ . Then we can conclude that

$$H^j(M_b; F) \xrightarrow{\sim} H^j(M_a; F)$$

for all  $j < k$ . □

## 6 Truncated Noncharacteristic Deformation

We will reformulate the noncharacteristic deformation lemma [KS90, Proposition 2.7.2] to make a relation to truncated microsupport.

**Proposition 6.1.** Let  $X$  be a Hausdorff space,  $F \in \mathbf{D}^b(\mathbf{Z}_X)$ , and let  $(U_t)_{t \in \mathbf{R}}$  be a family of open subsets of  $X$ . We assume the following conditions.

- (i)  $U_t = \bigcup_{s < t} U_s$ , for all  $t \in \mathbf{R}$ .
- (ii) For all pairs  $(s, t)$  with  $s \leq t$ , the set  $\overline{U_t - U_s} \cap \text{supp } F$  is compact.
- (iii) Setting  $Z_s := \bigcap_{t > s} \overline{U_t - U_s}$ , we have for all pairs  $(s, t)$  with  $s \leq t$ , and all  $x \in Z_s - U_t$ ,

$$(H_{X - U_t}^j(F))_x = 0 \quad \text{for all } j \leq k.$$

- (iv) For any  $s \in \mathbf{R}$ ,  $\bigcap_{t > s} U_t = \overline{U_s}$  and  $\overline{U_s}$  is closed in a paracompact open subset of  $X$ .
- (v) <sup>\*6</sup>  $Z_s - U_s \subset \partial U_s$  for all  $s \in \mathbf{R}$ .

Then, for all  $t \in \mathbf{R}$ , we have the isomorphisms

$$H^j \left( \bigcup_s U_s; F \right) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for all } j < k,$$

and the injections

$$H^k \left( \bigcup_s U_s; F \right) \hookrightarrow H^k(U_t; F).$$

**Proof.** Consider the following conditions for  $j \in \mathbf{Z}$ ,  $s$ , and  $t$ :

$$\begin{aligned} (a)_s^j & \quad \mu_s^j: \varinjlim_{t > s} H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F), \\ (a')_s^k & \quad \mu_s^k: \varinjlim_{t > s} H^k(U_t; F) \hookrightarrow H^k(U_s; F), \\ (b)_s^j & \quad \lambda_s^j: H^j(U_t; F) \xrightarrow{\sim} \varprojlim_{s < t} H^j(U_s; F). \end{aligned}$$

Let us prove  $(a)_s^j$  for  $s \in \mathbf{R}$  and  $j < k$ . We can assume that  $\overline{U_t - U_s}$  is compact for any  $s \leq t$  by replacing  $X$  by  $\text{supp } F$ . Consider the distinguished triangle

$$\mathbf{R}\Gamma_{X - U_t}(F) \rightarrow F \rightarrow \mathbf{R}\Gamma_{U_t}(F) \xrightarrow{+1}.$$

---

<sup>\*6</sup>This condition might be deduced from (iii).



Applying to this triangle the functor  $(\cdot)_{\overline{U}_t}$ , we have

$$(\mathrm{R}\Gamma_{X \dashv U_t}(F))_{\overline{U}_t} \rightarrow F_{\overline{U}_t} \rightarrow \mathrm{R}\Gamma_{U_t}(F) \xrightarrow{+1}.$$

Again applying to this triangle the functor  $\mathrm{R}\Gamma(X; \cdot)$ , we have

$$(6.1) \quad \mathrm{R}\Gamma(X; (\mathrm{R}\Gamma_{X \dashv U_t}(F))_{\overline{U}_t}) \rightarrow \mathrm{R}\Gamma(\overline{U}_t; F) \rightarrow \mathrm{R}\Gamma(U_t; F) \xrightarrow{+1}.$$

By hypotheses (iii)–(v), it follows that

$$\left( H_{X \dashv U_t}^j(F) \right)_{\partial U_t} = 0$$

for any  $j \leq k$  and then

$$H^j(X; (\mathrm{R}\Gamma_{X \dashv U_t}(F))_{\overline{U}_t}) = 0 \quad \text{for any } j \leq k.$$

<sup>\*7</sup>Thus for any  $t$ , we have

$$(6.2) \quad H^j(\overline{U}_t; F) \xrightarrow{\sim} H^j(U_t; F) \quad \text{for any } j < k,$$

$$(6.3) \quad H^k(\overline{U}_t; F) \hookrightarrow H^k(U_t; F).$$

Therefore, we have by (iv) and (6.2),

$$\begin{aligned} \varinjlim_{t>s} H^j(U_t; F) &\cong H^j\left(\bigcap_{t>s} U_t; F\right) \\ &= H^j(\overline{U}_s; F) \\ &\xrightarrow{\sim} H^j(U_s; F) \end{aligned}$$

for any  $j < k$ , and (a)<sub>s</sub><sup>j</sup> follows. Similarly, we have by (iv) and (6.3),

$$\begin{aligned} \varinjlim_{t>s} H^k(U_t; F) &\cong H^k\left(\bigcap_{t>s} U_t; F\right) \\ &= H^k(\overline{U}_s; F) \\ &\hookrightarrow H^k(U_s; F) \end{aligned}$$

and (a')<sub>s</sub><sup>k</sup> follows. □

**Remark 6.2.** Lemma 5.2 follows from Proposition 6.1. Indeed, putting  $X = ]-\infty, b[$ ,  $F = G|_{I_b}$ , and  $U_t = I_t$ , we find that

- (i)  $\bigcup_{s<t} I_s = \bigcup_{s<t} ]-\infty, s[ = ]-\infty, t[ = I_t$  for all  $t \in \mathbf{R}$ .
- (ii) If  $s \leq t$ , then  $\overline{I_t - I_s} \cap \mathrm{supp} G = \overline{[s, t[} \cap \mathrm{supp} G = [s, t] \cap \mathrm{supp} G$  is compact.
- (iii) For  $s \in \mathbf{R}$ , we have

$$Z_s = \bigcap_{t>s} \overline{I_t - I_s} = \bigcap_{t>s} \overline{[s, t[} = \bigcap_{t>s} [s, t] = \{s\},$$

and

$$Z_s - I_t = \{s\} - ]-\infty, t[ = \begin{cases} \{s\} & \text{if } s = t, \\ \emptyset & \text{if } s < t. \end{cases}$$

---

<sup>\*7</sup> この辺の議論はきちんと結めないといけない。

By hypothesis of Lemma 5.2, for all  $s \in \{s\}$ ,

$$H_{X \leftarrow I_s}^j(F)_s = H_{[s, \infty[}^j(F)_s = 0.$$

Then we can apply Proposition 6.1, and we have the isomorphisms for all  $t \in [a, b[$  and  $j < k$ ,

$$H^j(I_b; F) = H^j\left(\bigcup_s I_s; F\right) \xrightarrow{\sim} H^j(I_t; F)$$

and the injections

$$H^k(I_b; F) = H^k\left(\bigcup_s I_s; F\right) \hookrightarrow H^k(I_t; F)$$

## 7 Relative Microlocal Morse Theory via Truncated Microsupport

Let  $f: Y \rightarrow X$  be a morphism of manifolds, and  $\varphi$  a real valued function of class  $C^1$ . For a real number  $t \in \mathbf{R}$ , we set

$$\begin{aligned} Y_t &:= \{y \in Y ; \varphi(y) < t\} = \varphi^{-1}(]-\infty, t[), \\ \tilde{Y}_t &:= \{y \in Y ; \varphi(y) \leq t\} = \varphi^{-1}(]-\infty, t]). \end{aligned}$$

Let us define the following morphisms:

$$\begin{array}{ccc} Y_t & \xrightarrow{j_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X, \end{array} \quad \begin{array}{ccc} \tilde{Y}_t & \xrightarrow{\tilde{j}_t} & Y \\ & \searrow f_t & \downarrow f \\ & & X. \end{array}$$

**Proposition 7.1.** Let  $G \in D^b(Y)$ , and assume  $f_t: \text{supp}(G) \cap \overline{Y}_t \rightarrow X$  is proper for all  $t \in \mathbf{R}$ . Let  $k$  be an integer and  $t_o$  a real number.

(i) Assume that, for any  $y \in Y - \tilde{Y}_{t_o}$ , we have

$$d\varphi(y) \notin \left( \text{SS}_k(G) + f_d \left( Y \times_X T^*X \right) \right).$$

Then

$$\begin{aligned} \text{(a)} \quad & R^j f_* G \cong R^j f_{t*} j_t^{-1} G \quad \text{for } j < k \text{ and } t > t_o, \\ \text{(a}_k\text{)} \quad & R^k f_* G \hookrightarrow R^k f_{t*} j_t^{-1} G \quad \text{for } t > t_o, \\ \text{(a')} \quad & R^j \tilde{f}_* G \cong R^j \tilde{f}_{t*} \tilde{j}_t^{-1} G \quad \text{for } j < k \text{ and } t \geq t_o, \\ \text{(a}'_k\text{)} \quad & R^k \tilde{f}_* G \hookrightarrow R^k \tilde{f}_{t*} \tilde{j}_t^{-1} G \quad \text{for } t \geq t_o, \\ \text{(b)} \quad & \text{SS}_k(Rf_* G) \subset f_\pi \left( f_d^{-1} \left( \text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\ & = f_\pi f_d^{-1} (\text{SS}_k(G)). \end{aligned}$$

(ii) Assume that, for any  $y \in Y - \tilde{Y}_{t_o}$ , we have

$$-d\varphi(y) \notin \left( \text{SS}_k(G) + f_d \left( Y \times_X T^*X \right) \right).$$

Then

$$\begin{aligned}
(c) \quad & R^j f_! G \cong R^j f_{t!} j_t^{-1} G \quad \text{for } j < k \text{ and } t > t_o, \\
(c_k) \quad & R^k f_! G \hookrightarrow R^k f_{t!} j_t^{-1} G \quad \text{for } t > t_o, \\
(c') \quad & R^j \tilde{f}_! G \cong R^j \tilde{f}_{t*} \tilde{j}_t^! G \quad \text{for } j < k \text{ and } t \geq t_o, \\
(c'_k) \quad & R^k \tilde{f}_! G \hookrightarrow R^k \tilde{f}_{t*} \tilde{j}_t^! G \quad \text{for } t \geq t_o, \\
(d) \quad & \text{SS}_k(Rf_! G) \subset f_\pi \left( f_d^{-1} \left( \text{SS}_k(G) \cap \pi^{-1}(\tilde{Y}_{t_o}) \right) \right) \\
& \subset f_\pi f_d^{-1}(\text{SS}_k(G)).
\end{aligned}$$

**Proof.** Let  $\tilde{f}$  be the morphism  $(f, \varphi): Y \rightarrow X \times \mathbf{R}$ , and let  $q$  and  $\tilde{\varphi}$  be the first and second projections defined on  $X \times \mathbf{R}$  respectively. Then  $\tilde{f}$  is proper on  $\text{supp}(G)$  and  $f = q \circ \tilde{f}$ .<sup>\*8</sup> Let  $i_t$  be the open embedding  $X \times ]-\infty, t[ \hookrightarrow X \times \mathbf{R}$  and  $\tilde{i}_t$  the closed embedding  $X \times ]-\infty, t] \hookrightarrow X \times \mathbf{R}$ . Set  $q_t = q \circ i_t$  and  $\tilde{q}_t = q \circ \tilde{i}_t$ . These morphisms can be visualized by the following diagrams.

$$\begin{array}{ccc}
Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} & \xrightarrow{q} & X, \\
j_t \uparrow & & \uparrow i_t & \nearrow q_t & \\
Y_t & \xrightarrow{\tilde{f}_t} & X \times ]-\infty, t[ & & 
\end{array}
\quad
\begin{array}{ccc}
Y & \xrightarrow{\tilde{f}} & X \times \mathbf{R} & \xrightarrow{q} & X. \\
\tilde{j}_t \uparrow & & \uparrow \tilde{i}_t & \nearrow \tilde{q}_t & \\
Y_t & \xrightarrow{\tilde{f}_t} & X \times ]-\infty, t] & & 
\end{array}$$

Then, setting  $\tilde{G} := Rf_* G$ , we have

$$\begin{aligned}
Rf_* G &\cong Rq_* \tilde{G}, & R\tilde{f}_* G &\cong R\tilde{q}_* \tilde{G}, \\
Rf_{t*} j_t^{-1} G &\cong Rq_{t*} i_t^{-1} \tilde{G}, & R\tilde{f}_{t*} \tilde{j}_t^{-1} G &\cong R\tilde{q}_{t*} \tilde{i}_t^{-1} \tilde{G}, \\
Rf_{t!} j_t^{-1} G &\cong Rq_{t!} i_t^{-1} \tilde{G}, & R\tilde{f}_{t*} \tilde{j}_t^! G &\cong R\tilde{q}_{t*} \tilde{i}_t^! \tilde{G}.
\end{aligned}$$

Thus taking  $j$ -th cohomology, we have

$$\begin{aligned}
R^j f_* G &\cong R^j q_* \tilde{G}, & R^j \tilde{f}_* G &\cong R^j \tilde{q}_* \tilde{G}, \\
R^j f_{t*} j_t^{-1} G &\cong R^j q_{t*} i_t^{-1} \tilde{G}, & R^j \tilde{f}_{t*} \tilde{j}_t^{-1} G &\cong R^j \tilde{q}_{t*} \tilde{i}_t^{-1} \tilde{G}, \\
R^j f_{t!} j_t^{-1} G &\cong R^j q_{t!} i_t^{-1} \tilde{G}, & R^j \tilde{f}_{t*} \tilde{j}_t^! G &\cong R^j \tilde{q}_{t*} \tilde{i}_t^! \tilde{G}.
\end{aligned}$$

Therefore, it is enough to prove the result for  $\tilde{G}$ ,  $q$ ,  $i_t$ ,  $\tilde{\varphi}$  instead of  $G$ ,  $f$ ,  $j_t$ ,  $\varphi$ .

(i) It follows from the hypothesis that

$$(7.1) \quad \text{SS}_k(\tilde{G}) \cap (T^*X \times \{(t; \tau); t > t_o\}) \subset T^*X \times \{(t; \tau); \tau \leq 0\}.$$

Indeed, by hypothesis, for any  $(y; \eta) \in \text{SS}_k(G)$  and  $(f(x); \xi) \in Y \times_X T^*X$ , we have

$$(7.2) \quad d\varphi(y) \neq \eta + f^* \xi.$$

If  $(x, t; \xi, \tau)$  is a point of  $\text{SS}_k(\tilde{G}) \cap (T^*X \times \{(t; \tau); t > t_o\})$ , then, by the formula (3.2), there exists  $(y; \eta) \in \text{SS}_k(G)$  such that

$$\tilde{f}(y) = (x, t) \text{ and } \eta = \tilde{f}^*(\xi, \tau).$$

<sup>\*8</sup>This can be proved as follows. If  $K$  is a compact set of  $X \times \mathbf{R}$ ,  $\tilde{f}^{-1}(K)$  is compact.

For this  $\eta$ , we have

$$\eta = \tilde{f}^*(\xi, \tau) = f^*\xi + \varphi^*\tau = f^*\xi + \tau d\varphi(y).$$

If  $\tau > 0$ , we get

$$d\varphi(y) = \frac{1}{\tau}\eta - \frac{1}{\tau}f^*\xi$$

and this contradicts (7.2) because  $\text{SS}_k(G)$  is conic.

By (7.1), Let  $U$  be an open subset of  $X$  contained in a local chart, and convex in such a chart. By Proposition 6.1, we have isomorphisms for all  $t > t_o$  and  $j < k$

$$(7.3) \quad H^j(U \times \mathbf{R}; \tilde{G}) \xrightarrow{\sim} H^j(U \times ]-\infty, t[; \tilde{G})$$

and have an injection for all  $t > t_o$

$$(7.4) \quad H^k(U \times \mathbf{R}; \tilde{G}) \hookrightarrow H^k(U \times ]-\infty, t[; \tilde{G}).$$

This can be showed as follows.

This proves the result. □

**Remark 7.2.**  $(U \times I_t)$  is not a gentle family.

## 参考文献

- [GKS12] Stéphane Guillermou, Masaki Kashiwara, Pierre Schapira, *Sheaf Quantization of Hamiltonian Isotopies and Applications to Nondisplaceability Problems*, Duke. Math. J. 161, No. 2, pp.201–245, 2012.
- [KFS03] Masaki Kashiwara, Teresa Monteiro Fernandes, Pierre Schapira, *Truncated Microsupport and Holomorphic Solutions of D-modules*, Ann. Scient. Éc. Norm. Sup., série 4, tome 36, pp.583–399, 2003.
- [KS90] Masaki Kashiwara, Pierre Schapira, *Sheaves on Manifolds*, Grundlehren der Mathematischen Wissenschaften, 292, Springer, 1990.
- [ST92] Pierre Schapira, Nobuyuki Tose, *Morse Inequalities for  $\mathbf{R}$ -constructible Sheaves*, Adv. Math. 93, pp.1–8, 1992.